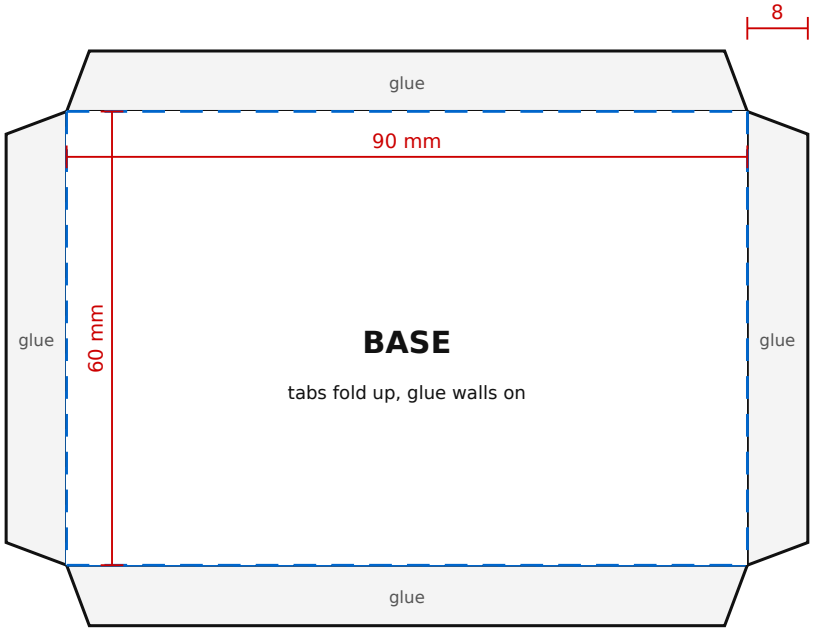
Robotics Project #13 · Templates 1/5 (1:1 scale)

Wheelbarrow Walker Robot

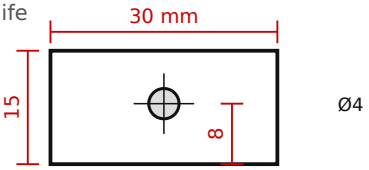
PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — — — fold (dashed) ◯ hole: grown-up pokes with a pencil/awl red = size in mm

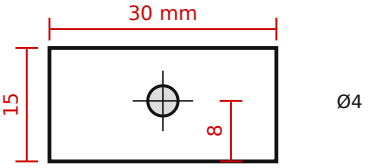
3 mm corrugated cardboard · grown-up cuts with craft knife



Barrow base 90x60 + 8 mm glue tabs (wheels roll freely)

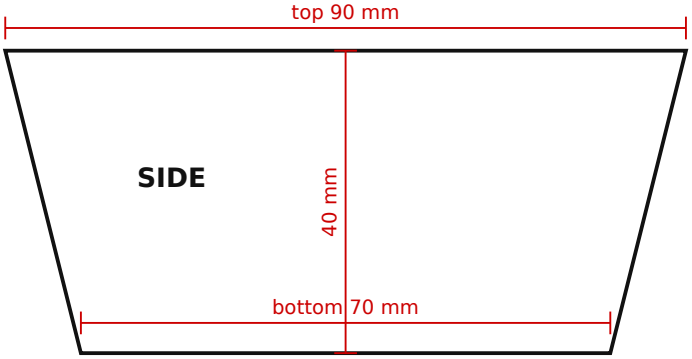


Axle hanger 1

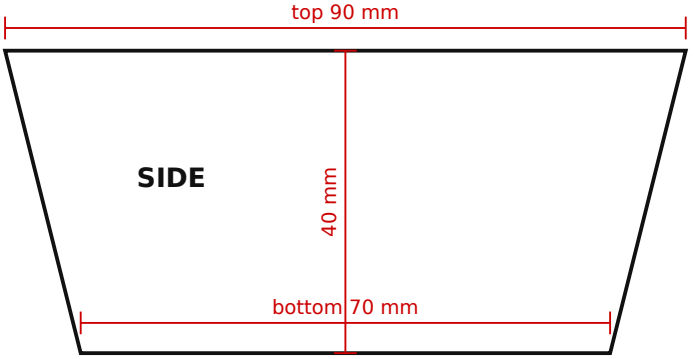


Axle hanger 2

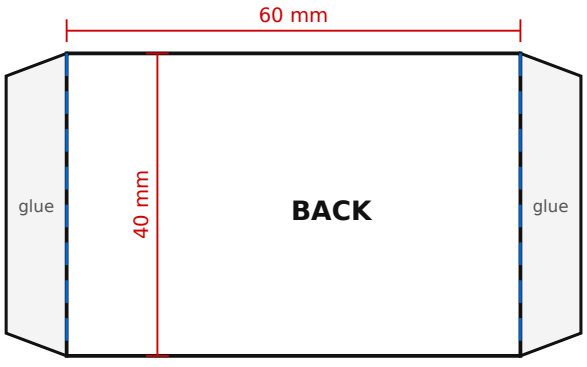
hole 8 mm up from bottom edge



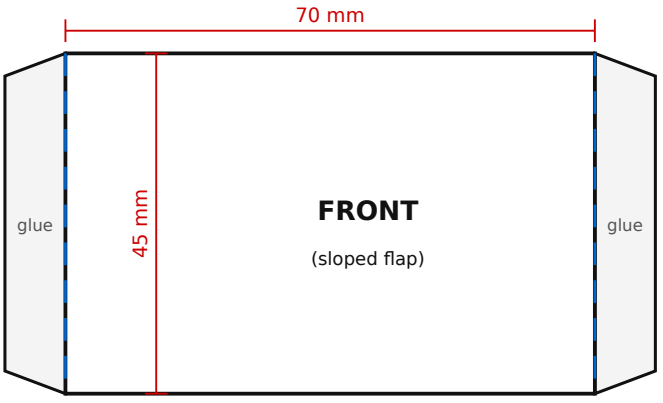
Side wall 1 (trapezoid)



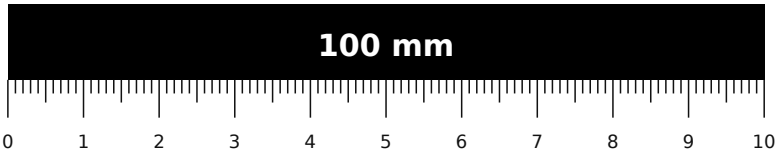
Side wall 2 (trapezoid)



Back wall 60x40 + tabs



Front flap 70x45 + tabs



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm.
If not, reprint at 100% (Actual size).
Use card ~1 mm (cereal box) for parts.

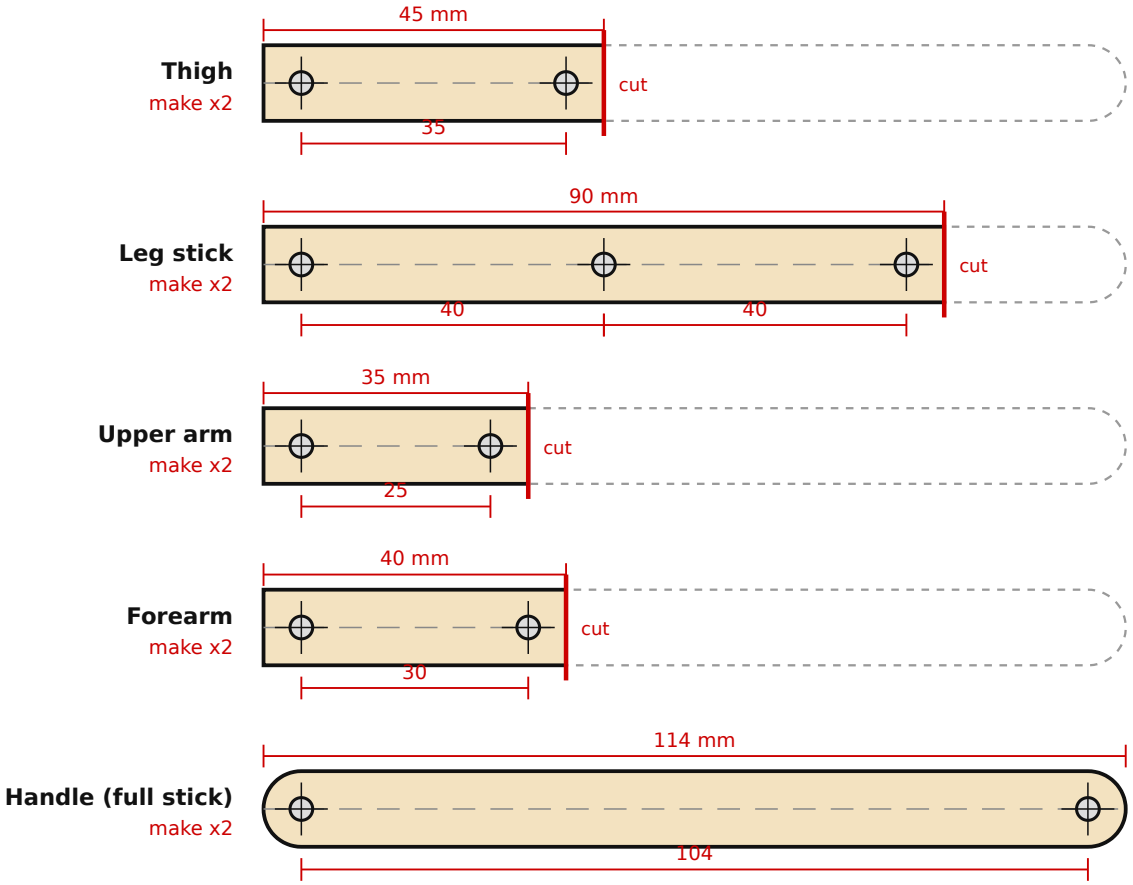
Robotics Project #13 · Templates 2/5 (1:1 scale)

Wheelbarrow Walker Robot

PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

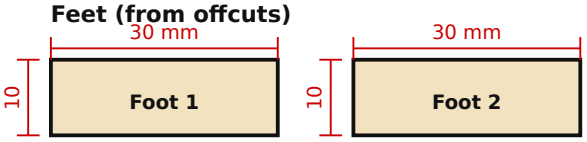
cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

Lay an ice-cream stick (114x10) on each guide, mark the cut at the red line, grown-up cuts.



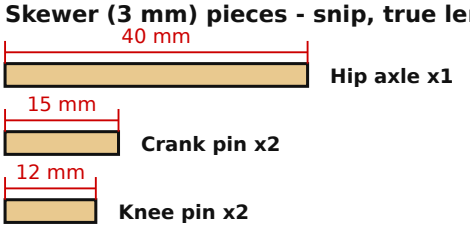
Ø3 pin holes 5 mm from the ends (leg stick: also one in the middle). Wiggle the pencil so pins turn freely.

Feet (from offcuts)



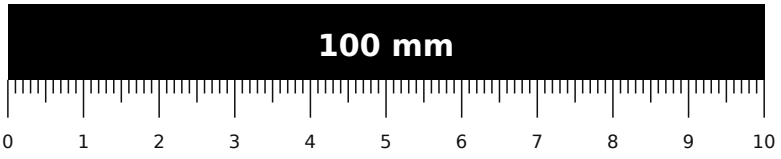
Glue the bottom end of each leg stick across a foot.
GRIP: wrap 2 rubber bands round each foot, or zig-zag lines of hot glue under it, so it pushes, not slides.

Skewer (3 mm) pieces - snip, true length:





Straw: 4 short rings (3 mm) slide on pin ends as stoppers; glue ring to pin, never to the stick.



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm.
If not, reprint at 100% (Actual size).
Use card ~1 mm (cereal box) for parts.

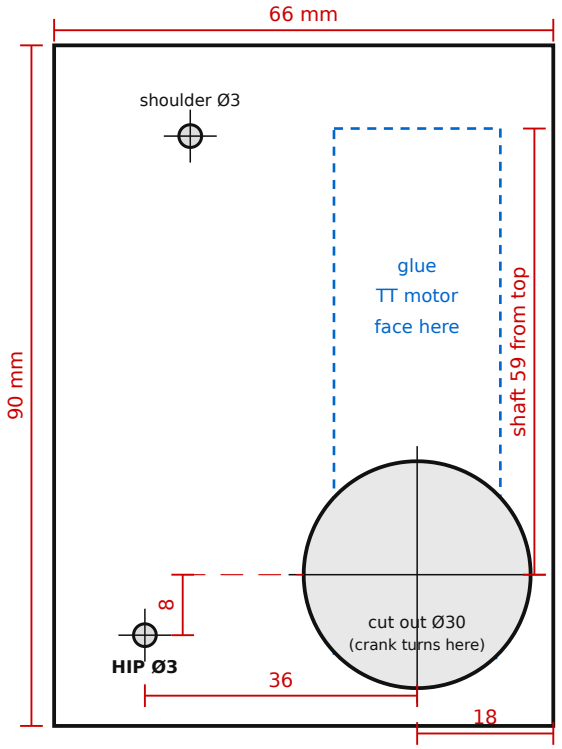
Robotics Project #13 · Templates 3/5 (1:1 scale)

Wheelbarrow Walker Robot

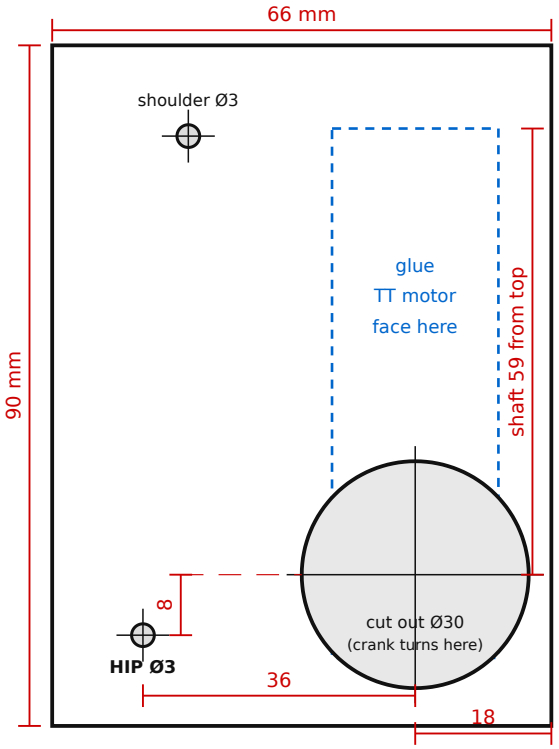
PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — — — fold (dashed) ○ hole: grown-up pokes with a pencil/awl red = size in mm

3 mm corrugated cardboard. Hip plates hold the TT motor upright in the hips; one crank on each shaft end.

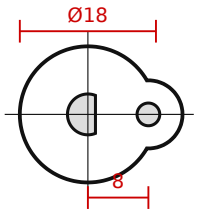


Hip plate LEFT
shaft through Ø30, motor upright

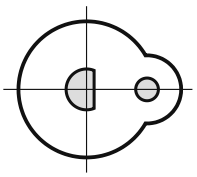


Hip plate RIGHT
same shape - glue it flipped over (mirror)

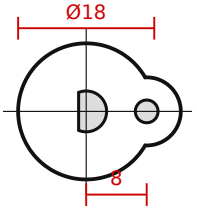
Hip hole: 36 mm behind and 8 mm below the shaft (keep exact - this sets the walk).



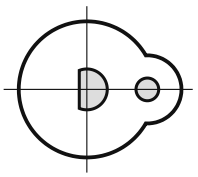
Crank disc A



Crank disc A



Crank disc B



Crank disc B

Glue the two A discs into one 6 mm crank, and the two B discs into another.

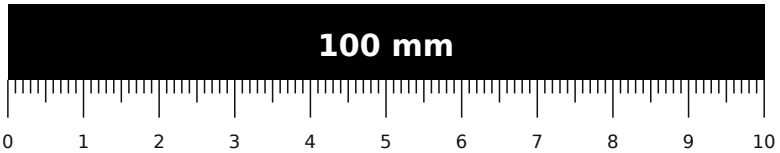
Centre: D-hole Ø5.4 with flat (3.7 across) - snug on the TT shaft, then a drop of hot glue.

A: flat on the lobe side. B: flat away from lobe. On the same shaft the flats line up,

so A and B pins point OPPOSITE ways = 180° apart = left, right, left, right!

Crank pin Ø3 at 8 mm; the lobe keeps 3 mm of card round the pin hole.

TT gear motor (double shaft): body ~70 x 22 x 18 mm, 5.4 mm D-shaft out both sides (~9 mm each).
Check yours: the Ø30 hole goes round the shaft, the dashed box shows where the body sits.





Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm.
If not, reprint at 100% (Actual size).
Use card ~1 mm (cereal box) for parts.

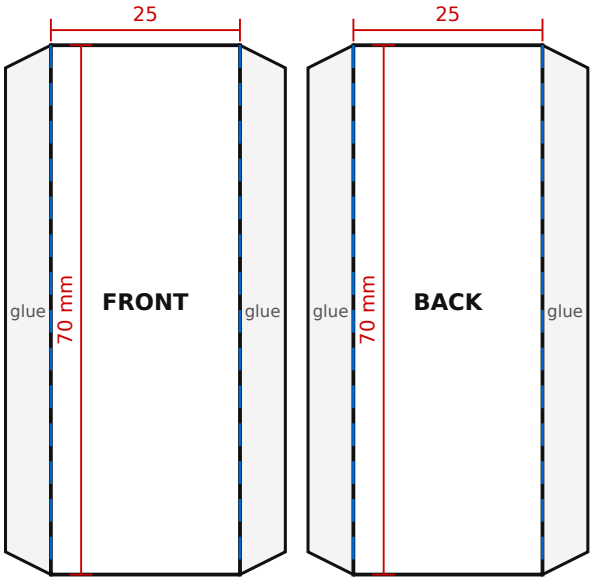
Robotics Project #13 · Templates 4/5 (1:1 scale)

Wheelbarrow Walker Robot

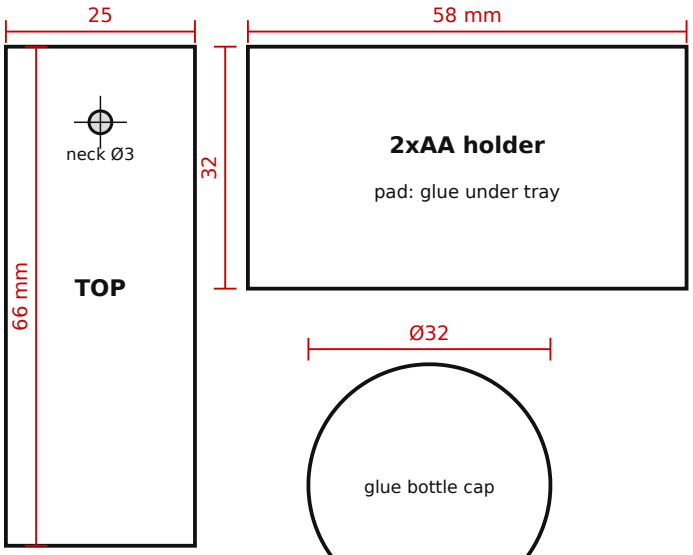
PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

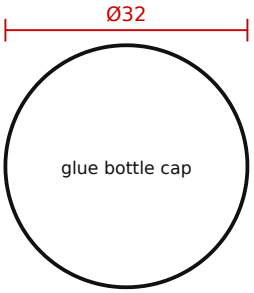
cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

Torso covers, battery pad, hat - and a 1:1 side view to check your leg before gluing.

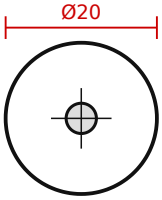


Torso covers 25x70 (+6 mm tabs glue to plate edges)

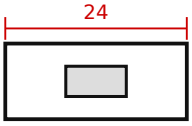




Hat brim



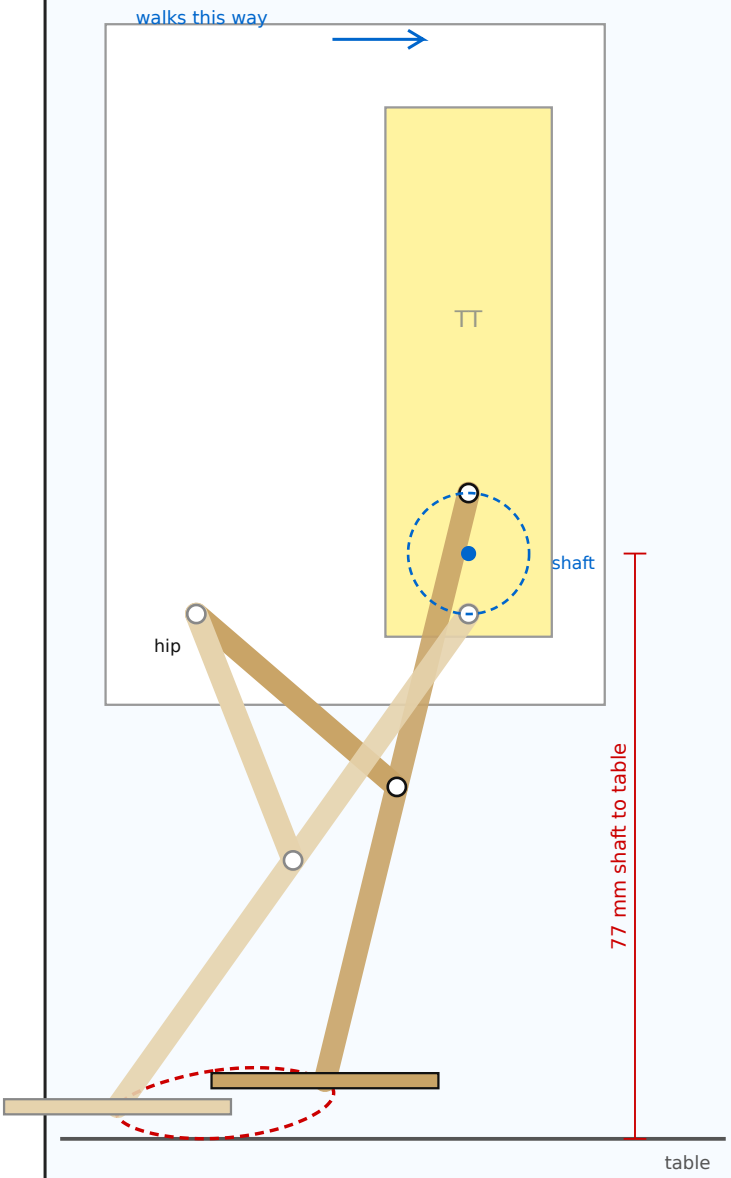
Wheel washer x2 (Ø4)



Switch plate (cut 8x4 slot)

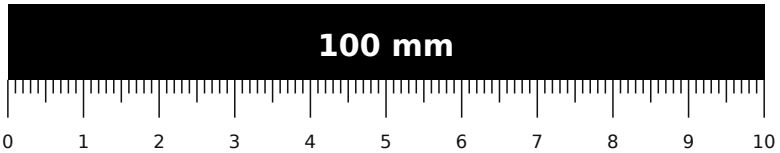
1:1 LEG CHECK (side view)

lay your leg on this drawing



Walks backwards? Swap the 2 motor wires.

Red dashes = path of the foot: flat bit ~22 mm on the table (the push), then it lifts ~9 mm and swings forward.



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm. If not, reprint at 100% (Actual size). Use card ~1 mm (cereal box) for parts.

github.com/dev-repos/kids-robotics-projects · Kids.JoinDev · CC BY 4.0 · Grown-ups do all craft-knife cutting.

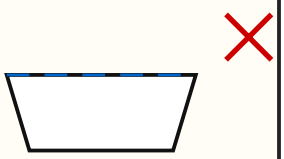
Robotics Project #13 · Templates 5/5 (1:1 scale)

Wheelbarrow Walker Robot

PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

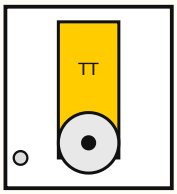
1 Cut and fold the barrow tray (sheet 1). Glue the tabs inside.



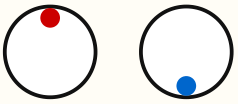
2 Skewer axle through a straw under the tray; glue a wheel on each end. Wheels roll freely.



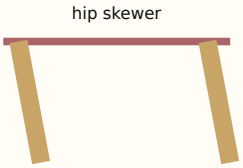
3 Hot-glue a hip plate (sheet 3) on each flat side of the TT motor, motor upright, shaft in the Ø30 hole.



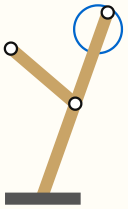
4 Glue crank A on one shaft end, crank B on the other: the pins must point opposite ways.



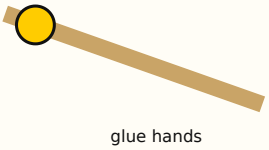
5 Push the 40 mm hip skewer through both hip holes. Put a thigh on each end, add straw stoppers.



6 Leg stick: top hole on the crank pin, middle hole pinned to the thigh knee, foot at the bottom.



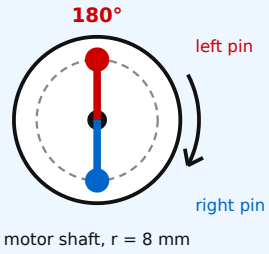
7 Glue torso covers, arms at the shoulders, hands onto the handles with both feet on the table.

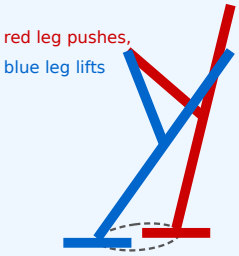


8 2xAA + switch under the tray, wires along a handle to the motor. Switch on and he walks!



How the legs walk





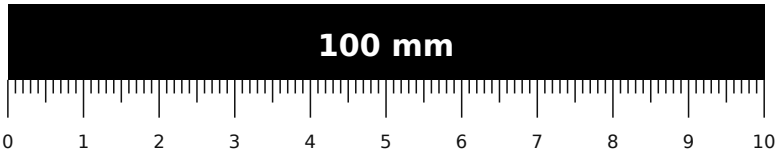
The motor turns both cranks. The pins are half a turn apart, so when the left foot pushes back on the table, the right foot lifts and swings forward - then they swap. Each push is ~22 mm; the foot lifts ~9 mm. The thigh is a guide: it holds the knee so the foot draws a flat-bottomed loop. His hands push the barrow along!

Parts checklist

- ☐ 3 mm cardboard
- ☐ Paper straw
- ☐ Rubber bands (feet)
- ☐ ~10 ice-cream sticks
- ☐ TT motor (double shaft)
- ☐ Ping-pong ball + cap
- ☐ 2 wheels (caps/65 mm)
- ☐ 2xAA holder + batteries
- ☐ Hot glue (grown-up)
- ☐ Skewers 3 mm
- ☐ Slide switch + wire
- ☐ Pencil, scissors

Safety first!

- A grown-up uses the hot glue gun, craft knife, and pokes the holes.
- Snip the sharp skewer tips off.
- Fingers away from the cranks when ON.
- Batteries the right way round; never join the wires; take them out after.
- Keep small parts from little kids.



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm. If not, reprint at 100% (Actual size). Use card ~1 mm (cereal box) for parts.

github.com/dev-repos/kids-robotics-projects · Kids.JoinDev · CC BY 4.0 · Grown-ups do all craft-knife cutting.